

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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FRUIT AND ITS COLOR USING BACKTRACKING

Aim:

To implement backtracking in Prolog to identify fruits and their colors.

Objective:

To demonstrate how Prolog searches through multiple possible solutions using backtracking.

Algorithm:

- 1. Start.**
- 2. Define fruit-color facts.**
- 3. Query a fruit or color.**
- 4. Prolog searches for a matching fact.**
- 5. If multiple matches exist, Prolog uses backtracking to find the next solution.**
- 6. Display all possible solutions.**
- 7. Stop.**

Flowchart:

START

|

v

Store Fruit-Color Facts

|

v

Enter Fruit/Color

|

v

Search Matching Fact

|

v

Match Found?

/ \

Yes No

| |

Display Fail

Solution

|

v

More Solutions?

/ \

Yes No

| |

Backtrack STOP

Prolog Program:

% Fruit and Color using Backtracking

fruit_color(apple, red).

fruit_color(apple, green).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(grapes, purple).

fruit_color(mango, yellow).

fruit_color(strawberry, red).

Query

?- fruit_color(apple, Color).

Output:

Color = red ;

Color = green.

Another query:

?- fruit_color(Fruit, green).

Output:

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93 ?- fruit_by_color(yellow, Fruit).  
Fruit = banana ▲
```

Result:

Thus, fruit and color information is successfully retrieved using Prolog backtracking.